Introduction to Econometrics
Notes on convergence modes and laws of large numbers

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Large sample theory

Consider a sequence $(z_1, z_2, \ldots, z_n)'$ of random variables denoted by $\{z_n\}$

Different modes of convergence:

- Convergence in probability
- Almost sure convergence
- Convergence in mean square
- Convergence to a random vector
- Convergence to a distribution

Large sample theory - some intuition

- It's important to realize that we are not talking about just a single variable, but an (infinitely long!) list of variables
- The ones later in the list are getting closer and closer to ... something
- This something might be a single number, or it might be an entire distribution
- To develop an intuition, we need to work out what "closer and closer" means
- The reason there are so many modes of convergence for random variables is that there are several types of "closeness" we might measure

Large sample theory - some intuition

- Let's say you have N balls in the box.
- You can pick them one by one.
- After you picked k balls, you can compute the mean weight of the balls in the box
- Your best answer would be $\bar{x_k} = \frac{1}{k}\sum_{i=1}^{k} x_i$.
- The $\bar{x}_k$ itself is the random value - it depends on which k balls you picked first.
- If you keep pulling the balls, at some point there will be no balls left in the box, and you'll get $\bar{x}_N \equiv \mu$
- Think of this box as containing infinite number of balls - if you could draw all of the balls, the above equality would hold
- We will see what happens as N gets sufficiently large

Large sample theory

Convergence in probability

- A sequence of random scalars $\{z_n\}$ **converges in probability** to a constant (nonrandom) α if, for any $\varepsilon > 0$,

$$\lim_{n\to\infty} Pr(|z_n - \alpha| > \varepsilon) = 0 \tag{1}$$

  The constant α is called the **probability limit** of z_n, also written as "$\underset{n\to\infty}{plim}\, z_n = \alpha$" or as "$z_n \underset{p}{\to} \alpha$" or as "$z_n - \alpha \underset{p}{\to} 0$".
- This definition of convergence in probability is extended to a sequence of random vectors or random matrices.
- That is, a sequence of K-dimensional random vectors $\{\mathrm{z}_n\}$ converges in probability to a K-dimensional vector of constants $\boldsymbol{\alpha}$ if, for any $\varepsilon > 0$,

$$\lim_{n\to\infty} Pr(|z_{nk} - \alpha_k| > \varepsilon) = 0, \text{ for all } k = 1, \ldots, K \tag{2}$$

Large sample theory

Example:

- Let $X_1, X_2, \ldots$ be Bernoulli random variables (each with its own probability of being a 1).
- A realization of this sequence is just a sequence of 1's and 0's.
- Unpacking the definitions, $X_n \underset{p}{\to} 0$ is equivalent to $P(X_n - 1) \to 0$
- That is, convergence to 0 in probability says that the 1's will get rarer and rarer as one looks ahead in the sequence.
- So convergence in probability says that any particular term with n large is likely to be close to the limit.

Large sample theory

Almost sure convergence

- A sequence of random scalars $\{z_n\}$ **converges almost surely** to a constant α if,

$$Pr(\lim_{n\to\infty} z_n = \alpha) = 1 \tag{3}$$

  We write this as "$z_n - \alpha \xrightarrow[a.s.]{} 0$"
- The extension random vectors (matrices) is obvious.
- The concept of almost sure convergence is stronger than convergence in probability:

$$z_n - \alpha \xrightarrow[a.s.]{} 0 \Rightarrow z_n - \alpha \xrightarrow[p]{} 0$$

Large sample theory

Example: consider the same sequence of Bernoulli random variables

- $X_n \underset{a.s}{\longrightarrow} 0$ is equivalent to statement that with probability 1, there exists N such that $X_n = 0$ for all $n \geq N$
- That is convergence to 0 almost surely says that the 1's will completely go extinct (with probability 1)
- Therefore almost sure convergence is stronger than convergence in probability as it says something about the entire tail of the sequence
- Recall that convergence in probability says that any particular term with n large is likely to be close to the limit.

Large sample theory

Convergence in mean square
The concept of mean-square convergence, or convergence in mean-square, is based on the following intuition: a random variables is "close to a constant" if the square of their difference is on average small.

A sequence of random scalars $\{z_n\}$ **converges in mean squares** to a constant α if,

$$\lim_{n\to\infty} E[(z_n - \alpha)^2] = 0 \tag{4}$$

We write this as "$z_n - \alpha \xrightarrow[m.s.]{} 0$"

- Convergence in mean square is based on limit calculations for the moments of the random variable.
- In general, probabilistic and almost sure convergence are used but in some cases, limit calculations for moments are easier to construct.

Large sample theory

Convergence to a random vector

In each of the definitions above, the limit can also be a random vector $\mathbf{z}$. We say that a sequence of K-dimensional random vector $\{\mathbf{z}_n\}$ converges to a K-dimensional random vector $\mathbf{z}$ if $\{\mathbf{z}_n - \mathbf{z}\}$ converges to 0:

$$\text{"}\ \mathbf{z}_n \underset{p}{\rightarrow} \mathbf{z}\text{" is the same as "}\ \mathbf{z}_n - \mathbf{z} \underset{p}{\rightarrow} \mathbf{0}\text{" or "}\ \mathbf{z}_n = \mathbf{z} + o_p(1)\text{"}$$

where $o_p(1)$ is some suitable random vector (here $\mathbf{z}_n - \mathbf{z}$) that converges to $\mathbf{0}$ in probability.

Likewise

$$\text{"}\ \mathbf{z}_n \xrightarrow[a.s.]{} \mathbf{z}\text{" is the same as "}\ \mathbf{z}_n - \mathbf{z} \xrightarrow[a.s.]{} \mathbf{0}\text{"}$$

and

$$\text{"}\ \mathbf{z}_n \xrightarrow[m.s.]{} \mathbf{z}\text{" is the same as "}\ \mathbf{z}_n - \mathbf{z} \xrightarrow[m.s.]{} \mathbf{0}\text{"}$$

Large sample theory

Convergence in distribution

- Let $\{z_n\}$ be a sequence of random scalars and F_n be the cumulative distribution function (c.d.f.) of z_n.
- We say that $\{z_n\}$ **converges in distribution** to a random scalar z if

$$\lim_{n\to\infty} F_n(z) = F(z)$$

  for all points z where $F(.)$ is continuous.
- We write "$z_n \underset{d}{\to} z$" or "$z_n \underset{L}{\to} z$" and call F the **asymptotic** (or **limiting** or **limit**) **distribution** of z_n.

Large sample theory

- Sometimes we write "$z_n \underset{d}{\rightarrow} F$" when the distribution F is well-known, e.g. "$z_n \underset{d}{\rightarrow} N(0,1)$"
- It can be shown that

$$"z_n \underset{p}{\rightarrow} z" \Rightarrow "z_n \underset{d}{\rightarrow} z" \tag{5}$$

If z is a constant

$$"z_n \underset{d}{\rightarrow} z" \Rightarrow "z_n \underset{p}{\rightarrow} z"$$

Large sample theory

- The extension to a sequence of random vectors is immediate: $\mathbf{z}_n \underset{d}{\rightarrow} \mathbf{z}$ if the joint c.d.f. F_n of the random vector $\mathbf{z}_n$ converges to the joint c.d.f. F of $\mathbf{z}$ at every continuity point of F.
- Note, however, that, unlike the other concepts of convergence, for convergence in distribution, element-by-element convergence does not necessarily mean convergence for the vector sequence. That is,

$$z_{nk} \underset{d}{\rightarrow} z_k, \;\; k = 1, \ldots, K \not\Rightarrow \mathbf{z}_n \underset{d}{\rightarrow} \mathbf{z}$$

Large sample theory

Convergence in distribution - very informal explanation

- Consider a sequence of random variables X_n, and a random variable Y.
- If you take a large number of samples of Y, and a large number of samples of X_n, for each n, then you would expect the samples from X_n to be more statistically similar to the samples of Y as n gets larger.
  The distribution of the variable X_n approaches the distribution of the variable Y as n approaches infinity.

Large sample theory

A common way to establish the connection between scalar convergence and vector convergence in distribution is captured by the **Multivariate Convergence in Distribution Theorem:**

Let $\{ \mathbf{z}_n \}$ be a sequence of K -dimensional random vectors. Then: " $\mathbf{z}_n \underset{d}{\rightarrow} \mathbf{z}$" $\Leftrightarrow$ "$\boldsymbol{\lambda}' \mathbf{z}_n \underset{d}{\rightarrow} \boldsymbol{\lambda}' \mathbf{z}$" for any K -dimensional vector $\boldsymbol{\lambda}$ of real numbers."

So the multivariate convergence in distribution $\mathbf{z}_n \underset{d}{\rightarrow} \mathbf{z}$ happens if and only if the univariate convergence in distribution $\boldsymbol{\lambda}' \mathbf{z}_n \underset{d}{\rightarrow} \boldsymbol{\lambda}' \mathbf{z}$ happens for every constant vector $\boldsymbol{\lambda}$

Relation among modes of convergence

Lemma 2.2 (relationship among the four modes of convergence):

(a) $\mathbf{z}_n \xrightarrow[m.s]{} \alpha \Rightarrow \mathbf{z}_n \underset{p}{\rightarrow} \alpha$. So $\mathbf{z}_n \xrightarrow[m.s]{} \mathbf{z} \Rightarrow \mathbf{z}_n \underset{p}{\rightarrow} \mathbf{z}$

(b) $\mathbf{z}_n \xrightarrow[a.s]{} \alpha \Rightarrow \mathbf{z}_n \underset{p}{\rightarrow} \alpha$. So $\mathbf{z}_n \xrightarrow[a.s]{} \mathbf{z} \Rightarrow \mathbf{z}_n \underset{p}{\rightarrow} \mathbf{z}$

(c) $\mathbf{z}_n \underset{p}{\rightarrow} \alpha \Leftrightarrow \mathbf{z}_n \underset{d}{\rightarrow} \alpha$

Relation among modes of convergence

Having defined the various modes of convergence, we can state two results essential for developing large -sample theory

Lemma 2.3 (preservation of convergence for continuous transformation):
Suppose $\mathbf{a}()$ is a vector-valued continuous function that does not depend on n.

(a) $\mathbf{z}_n \underset{p}{\rightarrow} \alpha \Rightarrow \mathbf{a}(\mathbf{z}_n) \underset{p}{\rightarrow} \mathbf{a}(\alpha)$, or stated differently

$$\underset{n\rightarrow\infty}{plim}\ \mathbf{a}(\mathbf{z}_n) = \mathbf{a}(\underset{n\rightarrow\infty}{plim}\ \mathbf{z}_n)$$

provided that $plim(\mathbf{z}_n)$ exists

(b) $\mathbf{z}_n \underset{d}{\rightarrow} \mathbf{z} \Rightarrow \mathbf{a}(\mathbf{z}_n) \underset{d}{\rightarrow} \mathbf{a}(\mathbf{z})$,

Relation among modes of convergence

An immediate implication of Lemma 2.3 is the usual arithmetic operations preserve convergence in probability.

For example:
$"x_n \underset{p}{\to} \beta, y_n \underset{p}{\to} \gamma" \Rightarrow "x_n + y_n \underset{p}{\to} \beta + \gamma"$
$"x_n \underset{p}{\to} \beta, y_n \underset{p}{\to} \gamma" \Rightarrow "x_n y_n \underset{p}{\to} \beta\gamma"$

Relation among modes of convergence

The following result about combinations of convergence in probability and in distribution will be used repeatedly to derive asymptotic distribution of the estimator

Lemma 2.4 (Parts (a) and (c) are called the Slutzky's theorem)

(a) $\mathbf{x}_n \underset{d}{\rightarrow} \mathbf{x}, \ \mathbf{y}_n \underset{p}{\rightarrow} \boldsymbol{\alpha} \Rightarrow \mathbf{x}_n + \mathbf{y}_n \underset{d}{\rightarrow} \mathbf{x} + \boldsymbol{\alpha}$

(b) $\mathbf{x}_n \underset{d}{\rightarrow} \mathbf{x}, \ \mathbf{y}_n \underset{p}{\rightarrow} \mathbf{0} \Rightarrow \mathbf{y}_n' \mathbf{x}_n \underset{p}{\rightarrow} 0$

(c) $\mathbf{x}_n \underset{d}{\rightarrow} \mathbf{x}, \ \mathbf{A}_n \underset{p}{\rightarrow} \mathbf{A} \Rightarrow \mathbf{A}_n \mathbf{x}_n \underset{d}{\rightarrow} \mathbf{A}\,\mathbf{x}$ provided that matrix $\mathbf{A}_n$ and vector $\mathbf{x}_n$ are conformable. In particular, if $\mathbf{x} \sim N(\mathbf{0}, \Sigma)$, then $\mathbf{A}_n \mathbf{x}_n \underset{d}{\rightarrow} N(\mathbf{0}, \mathbf{A}\Sigma\mathbf{A}')$

(d) $\mathbf{x}_n \underset{d}{\rightarrow} \mathbf{x}, \ \mathbf{A}_n \underset{p}{\rightarrow} \mathbf{A} \Rightarrow \mathbf{x}_n' \mathbf{A}_n^{-1} \mathbf{x}_n \underset{d}{\rightarrow} \mathbf{x}' \mathbf{A}^{-1} \mathbf{x}$

Relation among modes of convergence

- By setting $\boldsymbol{\alpha} = \mathbf{0}$, part (a) implies:

$$\mathbf{x}_n \underset{d}{\rightarrow} \mathbf{x}, \ \mathbf{y}_n \underset{p}{\rightarrow} \mathbf{0} \Rightarrow \mathbf{z}_n = \mathbf{x}_n + \mathbf{y}_n \underset{d}{\rightarrow} \mathbf{x} \tag{6}$$

- We say that the two sequences $\mathbf{z}_n$ and $\mathbf{x}_n$ are **asymptotically equivalent**: when $\mathbf{y}_n = \mathbf{z}_n - \mathbf{x}_n \underset{p}{\rightarrow} \mathbf{0}$ then $\mathbf{z}_n \underset{a}{\sim} \mathbf{x}_n$ or $\mathbf{z}_n = \mathbf{x}_n + o_p(1)$

Relation among modes of convergence

- A standard trick in deriving the asymptotic distribution of a sequence of random variables is to find an asymptotically equivalent sequence whose asymptotic distribution is easier to derive.
- In particular, by replacing $\mathbf{y}_n$ by $\mathbf{y}_n - \boldsymbol{\alpha}$ in part (b) of lemma 2.4, we obtain

$$\mathbf{x}_n \underset{d}{\rightarrow} \mathbf{x},\ \mathbf{y}_n \underset{p}{\rightarrow} \boldsymbol{\alpha} \Rightarrow \mathbf{y}_n' \mathbf{x}_n \underset{a}{\sim} \boldsymbol{\alpha}' \mathbf{x}_n \text{ or } \mathbf{y}_n' \mathbf{x}_n = \boldsymbol{\alpha}' \mathbf{x}_n + o_p(1) \quad (7)$$

- Therefore, replacing $\mathbf{y}_n$ by its probability limit does not change the asymptotic distribution of $\mathbf{y}_n' \mathbf{x}_n$, provided $\mathbf{x}_n$ converges in distribution to some random vector $\mathbf{x}$.

Viewing Estimators as Sequences of Random Variables

- The most frequently used versions of convergence are **convergence in probability** and **convergence in distributions** (cf. lemma's 2.3 and 2.4)
- Let $\hat{\boldsymbol{\theta}}_n$ be an estimator of $\boldsymbol{\theta}$ based on a sample of size n.
- The sequence $\{\hat{\boldsymbol{\theta}}_n\}$ is an example of a sequence of random variables, so the concepts introduced for sequeces (e.g. convergence in probability) are applicable to $\{\hat{\boldsymbol{\theta}}_n\}$.
- We say that $\{\hat{\boldsymbol{\theta}}_n\}$ is **consistent** for $\boldsymbol{\theta}$ if $\underset{n\to\infty}{plim}\, \hat{\boldsymbol{\theta}}_n = \boldsymbol{\theta}$
- The **asymptotic bias** of $\hat{\boldsymbol{\theta}}_n$ is defined as $\underset{n\to\infty}{plim}\, \hat{\boldsymbol{\theta}}_n - \boldsymbol{\theta}$.
- So the estimator $\hat{\boldsymbol{\theta}}_n$ is consistent of the asymptotic bias is zero.

Viewing Estimators as Sequences of Random Variables

- For example, if $\hat{\theta}_n$ is an unbiased estimator for the parameter θ, and $Var(\hat{\theta}_n) \to \mathbf{0}$, then

$$\lim_{n\to\infty} E[(\hat{\theta}_n - \theta)^2] = 0$$

  , i.e. $\hat{\theta}_n \xrightarrow[m.s.]{} \theta$. Consequently, $\hat{\theta}_n \underset{p}{\to} \theta$ (lemma 2.2a). So in that case, $\hat{\theta}_n$ is a consistent estimator for θ.
- A consistent estimator $\hat{\boldsymbol{\theta}}_n$ is **asymptotically normal** if

$$\sqrt{n}(\hat{\boldsymbol{\theta}}_n - \boldsymbol{\theta}) \underset{d}{\to} N(\mathbf{0}, \Sigma)$$

  Such an estimator is called $\sqrt{n}$-**consistent**.
- The variance matrix Σ is called the **asymptotic variance** $Avar(\hat{\boldsymbol{\theta}}_n)$

Laws of Large Numbers and Central Limit Theorems

- For a sequence of random scalars $\{z_n\}$, the sample mean $\bar{z}_n$, is defined as
$$\bar{z}_n = \frac{1}{n}\sum_{i=1}^{n} z_i$$
- Consider the sequence $\{\bar{z}_n\}$.
- **Laws of large numbers** (LLNs) concern conditions under which $\{\bar{z}_n\}$ converges either in probability or almost surely.
- An LLN is called **strong** if the convergence is almost surely and **weak** if the convergence is in probability.

Laws of Large Numbers and Central Limit Theorems

Some intuition:
In probability theory, the law of large numbers (LLN) is a theorem that describes the result of performing the same experiment a large number of times.

According to the law, the average of the results obtained from a large number of trials should be close to the expected value, and will tend to become closer as more trials are performed.

The LLN is important because it "guarantees" stable long-term results for the averages of some random events.

Laws of Large Numbers and Central Limit Theorems

Recall part (a) of Lemma 2.2:

$\mathbf{z}_n \xrightarrow[m.s]{} \boldsymbol{\alpha} \Rightarrow \mathbf{z}_n \underset{p}{\rightarrow} \boldsymbol{\alpha}$. So $\mathbf{z}_n \xrightarrow[m.s]{} \mathbf{z} \Rightarrow \mathbf{z}_n \underset{p}{\rightarrow} \mathbf{z}$

It implies:

A Version of Chebychev's Weak LLN:

$$\lim_{n\to\infty} E(\bar{z}_n) = \mu, \lim_{n\to\infty} Var(\bar{z}_n) = 0 \Rightarrow \bar{z}_n \xrightarrow[m.s.]{} \mu$$

Moreover, lemma 2.2.a implies that

$$\bar{z}_n \xrightarrow[m.s.]{} \mu \Rightarrow \bar{z}_n \underset{p}{\rightarrow} \mu$$

Laws of Large Numbers and Central Limit Theorems

The following strong LLN assumes that $\{z_i\}$ is i.i.d. (independently and identically distributed), but the variance does not need to be finite.

Kolmogorov's Second Strong Law of Large Numbers: Let $\{z_i\}$ be i.i.d. with $E(z_i) = \mu$. Then $\bar{z}_n \xrightarrow[a.s.]{} \mu$.

Moreover, lemma 2.2.b implies that $\bar{z}_n \underset{p}{\rightarrow} \mu$

These LLNs extend readily to random vectors by requiring element-by-element convergence.

Central Limit Theorems

- Central Limit Theorems (CLTs) are about the limiting behavior of the difference between $\bar{\mathbf{z}}_n$, and $E(\bar{\mathbf{z}}_n)$ (which equals $E(\mathbf{z}_i)$ if $\{\mathbf{z}_i\}$ is i.i.d.) blown up by $\sqrt{n}$.
- The only Central Limit Theorem we need for the case of i.i.d. sequences is:

  Lindeberg-Levy CLT: Let $\{\mathbf{z}_i\}$ be i.i.d. with $E(\mathbf{z}_i) = \boldsymbol{\mu}$ and $Var(\mathbf{z}_i) = \Sigma$. Then

$$\sqrt{n}(\bar{\mathbf{z}}_n - \boldsymbol{\mu}) = \frac{1}{\sqrt{n}} \sum_{i=1}^{n} (\mathbf{z}_i - \boldsymbol{\mu}) \underset{d}{\rightarrow} N(\ \mathbf{0}, \Sigma)$$

- The usefulness of this theorem is that the distribution of $\sqrt{n}(\bar{\mathbf{z}}_n - \boldsymbol{\mu})$ approaches normality regardless of the shape of the distribution of the individual z_i

Central Limit Theorems

- Usually, the Lindeberg-Levy CLT is for a sequence of scalar random variables. The vector version displayed above is derived from the scalar version as follows.
- Let $\boldsymbol{\lambda}$ be any real vector of the same dimension as $\mathbf{z}_i$. Then $\{\boldsymbol{\lambda}' \mathbf{z}_i\}$ is a sequence of scalar random variables with $E(\boldsymbol{\lambda}' \mathbf{z}_i) = \boldsymbol{\lambda}'\boldsymbol{\mu}$ and $Var(\boldsymbol{\lambda}' \mathbf{z}_i) = \boldsymbol{\lambda}'\Sigma\boldsymbol{\lambda}$.
- The scalar version of Lindeberg-Levy then implies that:

$$\sqrt{n}(\boldsymbol{\lambda}' \bar{\mathbf{z}}_n - \boldsymbol{\lambda}'\boldsymbol{\mu}) = \boldsymbol{\lambda}'\sqrt{n}(\bar{\mathbf{z}}_n - \boldsymbol{\mu}) \underset{d}{\rightarrow} N(\mathbf{0}, \boldsymbol{\lambda}'\Sigma\boldsymbol{\lambda})$$

- But this limit distribution is the distribution of $\boldsymbol{\lambda}' \mathbf{x}$ where $\mathbf{x} \sim N(\mathbf{0}, \Sigma)$.
- So by the Multivariate Convergence in Distribution Theorem, $[\sqrt{n}(\mathbf{z}_n - \boldsymbol{\mu})] \underset{d}{\rightarrow} \mathbf{x}$

Central Limit Theorems

Intuition behind CLT?

- None...
- Someone once said the Central Limit Theorem defies intuition - we can give all sorts of stories, and we can do lots of software demonstrations, but the CLT is really mysterious. It's truly magical that the limiting distribution is Gaussian.
- ...One can think if the CLT is part of nature - It's just like gravity in that regard.
- ...The CLT was not invented by humans, although it was for sure proved by humans.

Literature

- Hayashi, chapter 2: 2.1 (except Lemma 2.5)